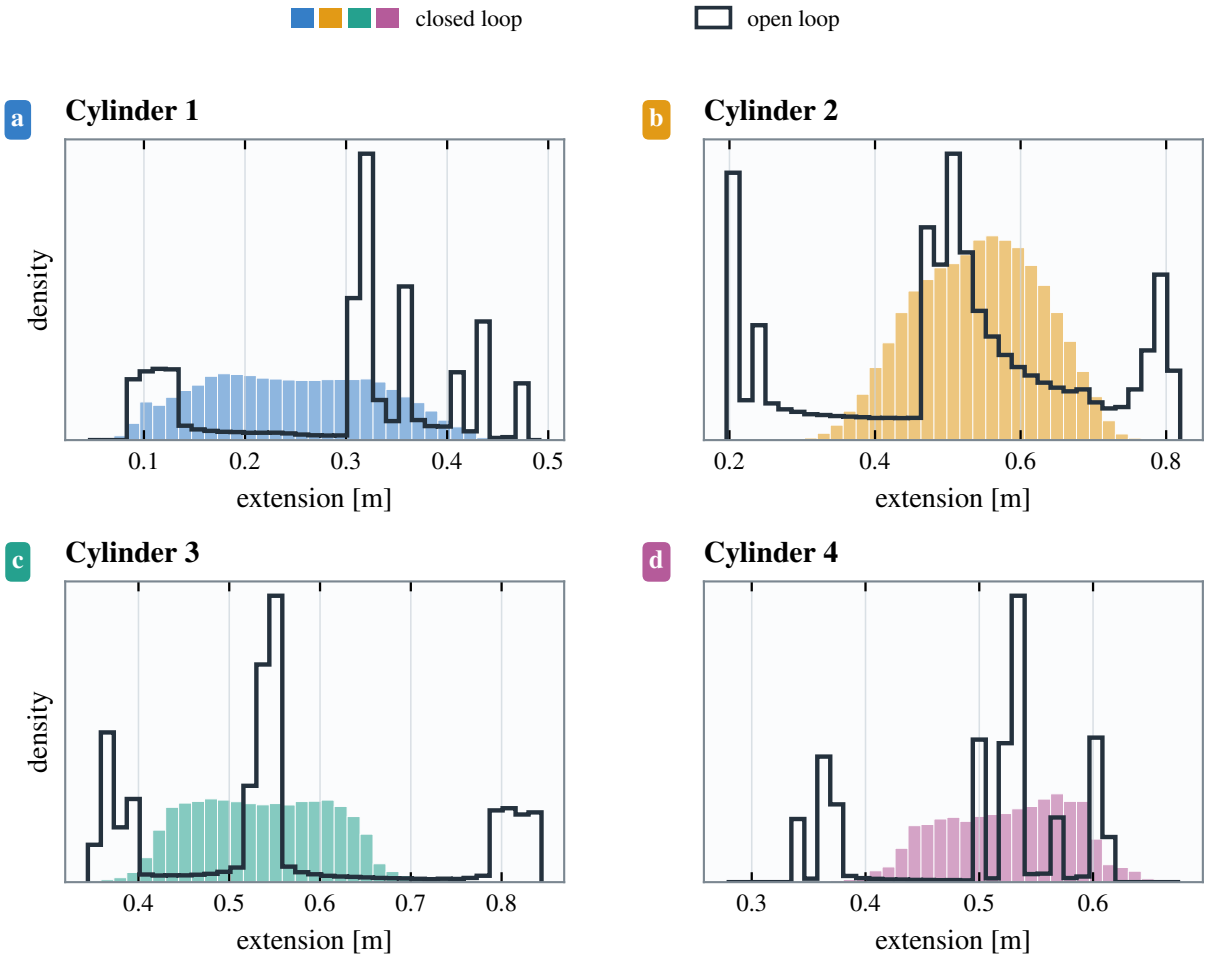


Figure 4.4: Extension coverage of the two campaigns, per cylinder and pooled over runs (closed-loop filled, open-loop outline). The closed-loop data fills the reachable range smoothly the open-loop data concentrates into the plateaus where the valve is identifiable.

at deployment, and neither campaign covers both. The forward model is therefore trained on their union.

The reduced-order model consumes the closed-loop material in a joined form. The inverse and forward exports are aligned on exact run identifiers, time indices, and sample times and the join is accepted only when the sixteen-dimensional cylinder state agrees between them sample for sample. The resulting command-conditioned dataset contains 452 400 transitions at 0.05 s drawn from 1 040 aligned runs. The logged valve command is carried through this join, but only as a supervision target for the model's internal valve head. It is never a runtime input, which is the property that lets the ROM be driven by desired cylinder motion alone (Section 5.4).[1]

Two closed-loop bodies of Simscape material therefore exist, they are easy to confuse, and only one of them trains anything in this thesis. The distinction is worth making sharply.

The *aligned export* is the one that trains the inverse, forward and reduced-order models. It is a nominal-plant closed-loop collection of 1 040 runs yielding 452 400 aligned transitions at 0.05 s, split by complete run into 936 training, 52 validation and 52 test. It predates